

5) Register and counter-Assignment

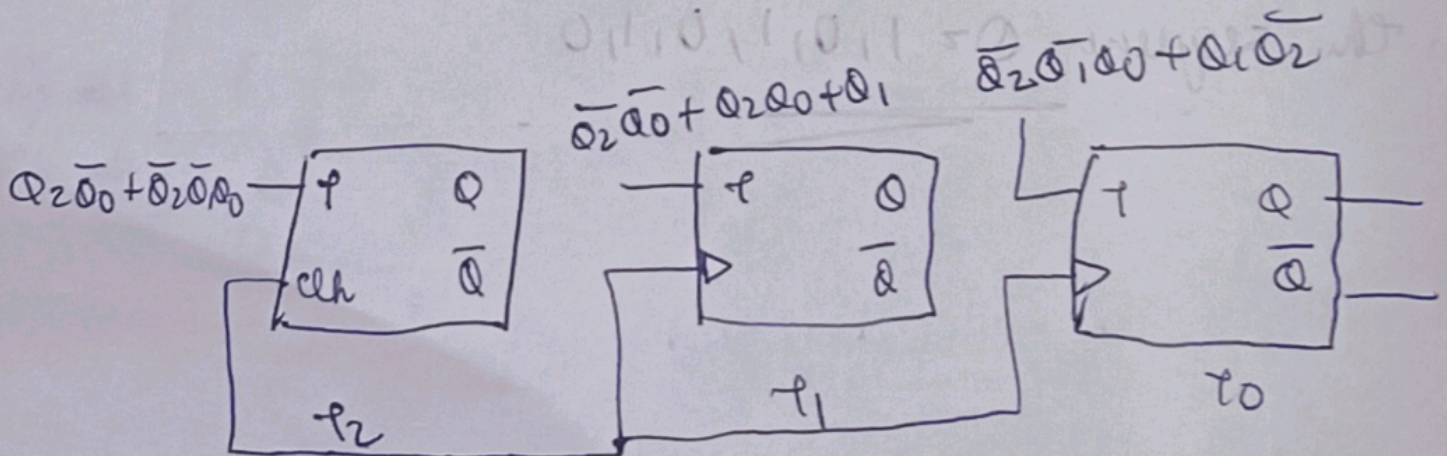
1) Design a counter with 4 flip-flop whose count sequence is 0, 1, 2, 4, 5, 3, 8, 0 - - -

Current state			next state			requirement		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	0	1	0
0	1	1	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
1	1	0	X	X	X	X	X	X
1	1	1	X	X	X	1	1	0

$$T_2 = Q_2 \bar{Q}_0 + \bar{Q}_2 \bar{Q}_1 Q_0$$

$$T_1 = \bar{Q}_2 \bar{Q}_0 + Q_2 Q_0 + Q_1$$

$$T_0 = \bar{Q}_2 \bar{Q}_1 Q_0 + Q_2 \bar{Q}_0$$



2) Design a 3 bit up/down counter using negative edge triggered T-FF

Input	Present state			next state			+ flip flop		
M	Q ₂	Q ₁	Q ₀	Q ₂ ⁺	Q ₁ ⁺	Q ₀ ⁺	T ₂	T ₁	T ₀
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	0	0	1	1	1	1
1	0	0	1	0	1	1	1	1	1
1	0	1	0	1	1	0	0	0	1
1	0	1	1	1	0	1	0	1	1
1	1	0	0	1	1	0	0	0	1
1	1	0	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0	1
1	1	1	1	0	0	1	0	0	1

$T_0 = 1$

MQ ₂	Q ₁ Q ₀ 00	01	10	11
00	0	1	1	0
01	0	1	1	0
10	1	0	0	1
11	1	0	0	1

$$T_1 = \bar{M}Q_0 + M\bar{Q}_0$$

0100

00 01 10 11

00

01

10

11

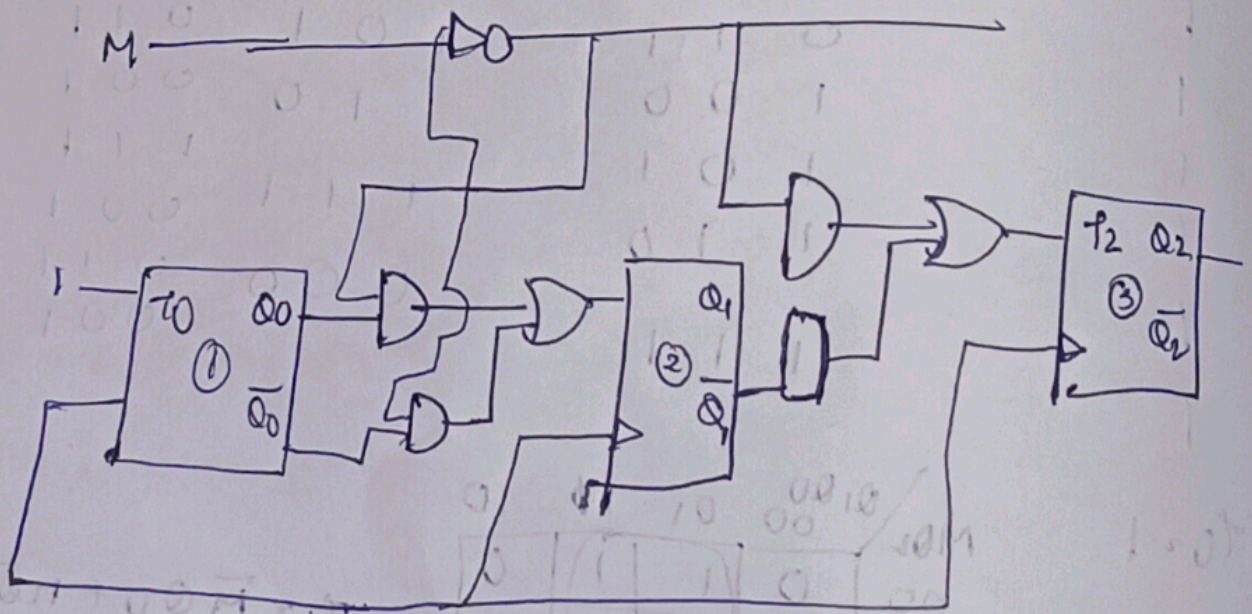
00	0	0	1	0
01	0	0	1	0
10	1	0	0	0
11	1	0	0	0

$$T_2 = \overline{M} Q_1 Q_0 + M \overline{Q}_1 \overline{Q}_0$$

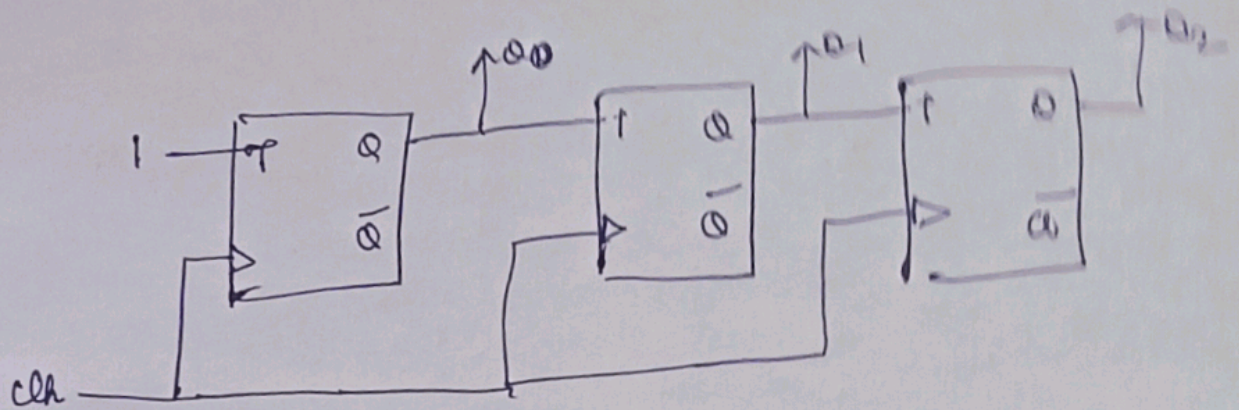
$$\therefore T_0 = 1$$

$$T_1 = \overline{M} Q_0 + M \overline{Q}_0$$

$$T_2 = \overline{M} Q_1 Q_0 + M \overline{Q}_1 \overline{Q}_0$$



3) The circuit in the figure below is counter what is the sequence that circuit counts in



There are negative edge triggered flip flop
assuming Initial state $Q_0, Q_1, Q_2 = 000$

sequence	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	0	0	0
5	0	0	1
6	0	1	0

$\therefore$ the sequence it follows $000 \rightarrow 001 \rightarrow 010 \rightarrow 111 \rightarrow 000$